BRAC University
Department of Computer Science and Engineering
CSE 437: Data Science—Coding With Real-World Data

A. **Course General Information:**

Course Code:	CSE437
Course Title:	Data Science: Coding With Real World Data
Credit Hours (Theory + Laboratory):	3 + 0
Contact Hours (Theory + Laboratory):	3 + 0
Category:	Program Elective
Type:	Elective, Engineering, Lecture
Prerequisites:	

B. **Course Catalog Description (Contents):**

Properties of data: variables and types, measurement scales, central tendency, dispersion, correlation, and contingency. Data distribution: probability, random variables, and probability distribution. Statistical inference: interval estimation, hypothesis testing, analysis of variance, and multivariate analysis of variance. Data preprocessing: data cleaning, integration, data transformation, and outlier detection. Feature engineering: data dimensionality reduction, component analysis, factor analysis, discriminant analysis, and feature selection. Time-series analysis: trend and seasonal components, stationary processes, non-stationary processes, forecasting methods, and advanced time-series analysis methods. Building models from data: predictive modeling through regression and sequential data models, classification modeling through supervised learning, and cluster modeling through unsupervised learning. Data visualization and model evaluation.

C. **Course Objective:**

The objective of this course is to describe both theoretical and practical approaches for making sense out of data. A step-by-step process is introduced, which is designed to walk you through the steps and issues that you will face in data analysis projects. It covers the more common tasks relating to the analysis of data including (1) how to prepare data prior to analysis, (2) how to generate summaries of the data, (3) how to identify non-trivial facts, patterns, and relationships in the data, and (4) how to create models from the data to better understand the data and make predictions. The hands-on exercise will help the students to apply data science methodology to solve a range of real-world problems.

D. **Course outcomes:**

After successful completion of this course, students will be able to:

Sl.	**CO Description**
CO1	**Apply** statistical inference methods for hypothesis testing and variance analysis of data.
CO2	**Use** data preprocessing and feature engineering methods for feature extraction and selection.
CO3	**Analyze** appropriate data modeling methods suitable for different types of problems and data
CO4	**Examine, select, and implement** proper data analytics steps for solving interesting real-life data science problems and write a report to show the results of data analysis.

E. **Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:**

Sl.	CO Description	POs	Bloom's Taxonomy Domain/Level	Delivery Methods, Activities	Assessment Tools
CO1	**Apply** statistical inference methods for hypothesis testing and variance analysis of data.	a	Cognitive/ Apply	Lecture	Assignment, Quiz, Exam
CO2	**Use** data preprocessing and feature engineering methods for feature extraction and selection.	a	Cognitive/ Apply	Lecture	Assignment, Quiz, Exam
CO3	**Evaluate** appropriate data modeling methods suitable for different types of problems and data	a	Cognitive/ Evaluate	Lecture	Assignment, Quiz, Exam
CO4	**Examine, select, and implement** proper data analytics steps for solving interesting real-life data science problems and write a report to show the results of data analysis.	c	Cognitive/ Create	Lab work	Assignment, Exam, Project

F. **Course Materials:**

Text and Reference Books:

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1.	Python Data Science Handbook: Essential Tools for Working with Data	Jake VanderPlas	2016	1st edition	O'REILLY	10: 1491912057
2.	The Elements of Statistical Learning: Data Mining, Inference and Prediction.	Trevor Hastie, Robert Tibshirani, Jerome Friedman	2009	2nd edition	Springer	10:0387848576

Other materials (if any)

- Lecture Notes and presentation slides
- DataCamp, AWS
- Video tutorials in buX

G. **Lesson Plan:**

Sl.	Topic	Week/ Lecture#	Related CO (if any)
1.	Introduction to Data Science Introduction to Data Science in DataCamp/Aws/ML	Week 1	CO1
2.	Data Preprocessing tasks such as Data Cleaning and Data Transformation (Normalization, Scaling), Dealing with missing values, MLE, EM	Week 2	CO2, CO1
3.	Data Mining: Association Rule Mining, FP-Growth	Week 3	CO1, CO4
4.	Feature Engineering Data Dimensionality Reduction (PCA, tSNE,	Week 4	CO2, CO4

	LDA), Feature Selection (mRMR, RFE)		
5.	Qualitative Data Analysis Methods Predictive Modeling using Regression Algorithms	Week 5	CO3, CO4
6.	Decision Theory: GINI Impurity and Decision Tree, CART, Regression Analysis	Week 6	CO3, CO4
	Mid-Term	Week 7	
7	Supervised Learning: Ada-Boost, SVM	Week 8	CO3, CO4
8	Unsupervised Learning (K-Means, K-Medoids)	Week 9	CO2, CO3
9.	Descriptive and Inferential Statistics	Week 10	CO2, CO1
10	Data Science in Natural Language Processing	Week 11	CO3, CO4
11.	Big Data processing platform: HADOOP, Map-Reduce/Spark Programming Paradigm	Week 12	CO3, CO4
12.	**Final Examination**	Week 13	

H. **Assessment Tools:**

Assessment Tools	**Weightage**
Attendance / Class Performance	05%
Quiz	10%
Midterm Exam	25%
Project $ Report	15%
Final Exam	30%
DataCamp/AWS Academy/Microsoft Learn Assignments	15%
Total=	100%

I. **CO Assessment Plan:**

Assessment Tools	**Course Outcomes**					
	CO1	**CO2**	**CO3**	**CO4**		
Assignments				x		
Quizzes and Class Performances	x	x	x			
Exam (Mid, Final)	x	x	x			
Project (report, presentation, demonstration)				x		

J. **CO Attainment Policy:** As per the course outcome attainment policy of the Department of Computer Science and Engineering

K. **Grading policy:** As per the BRAC University grading policy

L. **Course Coordinator: Dr. Md. Golam Rabiul Alam (GRA)**

M. **Additional information:**

Class Information:

Course ID: CSE437
Course Title: Data Science Coding With Real World Data
Sections: 01 - 04
Credits: 3

Semester: Summer 2026
Class Days: SAT, THURS: 2:00 pm - 3:20 pm
Class Hours: 3 hrs. per week
Class Room: [09C-16T]

Course Faculties:

Name: Dr. Md. Golam Rabiul Alam
Initial: GRA / MGR
Office: 4G23

Designation: Professor, CSE
Contact Number: +8801797347635 (Only in an emergency)
Email: rabiul.alam@bracu.ac.bd

Name: Md. Sabbir Ahmed
Initial: SBB
Office: 4M107

Designation: Senior Lecturer, CSE
Contact Number: + (Only in an emergency)
Email: sabbir.ahmed@bracu.ac.bd

Teaching-learning methodology:

- ✓ Classroom lectures
- ✓ Problem-solving and project development in DataCamp

A Few More Recommended Books:

1) Laura Igual, Santi Segui, **"Introduction to Data Science – A Python approach to concepts, techniques and applications"**, Springer.
2) Fracois Chollet, **"Deep Learning with Python"**, Manning Publications.
3) Josh Wills, Sandy Ryza, Sean Owen, and Uri Laserson, **"Advanced Analytics with Spark: Patterns for Learning from Data at Scale"**, O'Reilly.
5) Glenn J. Myatt, Wayne P. Johnson, **"Making Sense of Data I - A Practical Guide to Exploratory Data Analysis and Data Mining"**, 2nd edition, Wiley.
6) Hector Cuesta, Sampath Kumar, **"Practical Data Analysis"**.
7) Prabhanjan Tattar, Tony Ojeda, Sean Patrick Murphy, Benjamin Bengfort, Abhijit Dasgupta, **"Practical Data Science Cookbook - Data pre-processing, analysis and visualization using R and Python".**

Grading policy:

The grades at the University will be indicated in the following manner:

90 - 100 = A (4.0) Excellent
85 - <90 = A- (3.7)
80 - <85 = B+ (3.3)
75 - <80 = B (3.0) Good
70 - <75 = B- (2.7)
65 - <70 = C+ (2.3)
60 - <65 = C (2.0) Fair
57 - <60 = C- (1.7)
55 - <57 = D+ (1.3)
52 - <55 = D (1.0) Poor
50 - <52 = D- (0.7)
<50 = F (0.0) Failure

Grades without numerical value:

P: Pass

A course may be taken for a pass/fail grade providing that the instructor approves the option and the student carries 12 credits for regular letter grades in that semester.

I: Incomplete

Incomplete is assigned only when a student has failed to complete one or more course requirements for an unavoidable reason/accidental circumstance and has applied for an I grade. W: Withdrawal

Withdrawal is assigned to a student who withdraws from the course within the deadline for withdrawal with a 'W' grade.

Attendance policy:

The course has been designed to maintain linear dependencies for some topics. That is why the students are expected to maintain 100% attendance.

Gender policy:

Gender equity among male and female students in the class will be maintained as per the BRAC University's concern and BRAC's consistent endeavors on women's empowerment. Therefore, all students will be evaluated equally based on their performance in the course concerned, regardless of their gender.

Inclusive education policy statement:

Each of the students shall be given equal access to laboratory resources, relevant materials, and consultation hours, free from discrimination based on gender, language, sexual orientation, pregnancy, culture, ethnicity, religion, health or disability, socioeconomic background, or geographic location, as per the inclusive education policy of Bangladesh.

Course Repository:

Google drive:	**https://drive.google.com/drive/folders/1yuDdWMJmGDHLpIi8oOOXGQc5wFBMVCcG?usp=sharing**
Google Classroom	Class Code: TBA
buX:	**TBA**
DataCamp:	**TBA**
AWS Academy:	**TBA**
Microsoft Learn	**TBA**